JNTU Circle: Architectural Design, Real-Time Synchronization, and Multi-Regulation Evaluation for Engineering Educational Knowledge Systems

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ABSTRACT

Navigating undergraduate engineering coursework across institutions affiliated with Jawaharlal Nehru Technological University Anantapur (JNTUA) presents recurring operational hurdles. On our campus, concurrent regulatory frameworks--specifically R15, R19, R20, and R23--frequently cause students to revise from obsolete lecture materials, miss critical examination announcements during central server outages, and miscalculate semester GPAs due to shifting internal evaluation weights. To eliminate these recurring obstacles, our student research group at MS Engineering College developed JNTU Circle: a production-deployed academic knowledge hub, automated circular ingestion pipeline, and multi-scheme evaluation platform. Built with React 19, Express.js, and MongoDB Atlas, the architecture incorporates: (1) a multi-regulation digital repository categorizing validated study resources across 6 engineering disciplines; (2) a decoupled scraping daemon utilizing SHA-256 cryptographic DOM fingerprinting to broadcast genuine university circulars within 60 seconds without duplicates; (3) an in-browser PDF reverse proxy that bypasses cross-origin framing constraints for inline document inspection; (4) an adaptive grading algorithm calculating SGPA and CGPA under shifting continuous-evaluation splits; and (5) a curriculum-grounded conversational agent deployed on Groq LPU hardware delivering sub-200ms token streaming. Empirical trials conducted with N = 210 engineering scholars across two consecutive semesters demonstrated a 96.2% acceleration in finding verified study resources ($p < 0.001$) and an outstanding System Usability Scale (SUS) score of 89.4/100, verifying the practical efficacy of our platform.

Keywords: *Curricular Knowledge Hub, Real-Time Portal Ingestion, Multi-Regulation Evaluation, Engineering Informatics, Groq LPU Inference, PDF Streaming Pipeline, Progressive Web Application.*

1. INTRODUCTION

1.1 The Affiliating Collegiate Framework in Regional Technical Education

State-governed technical universities in India, including JNTUA, function through a federated affiliation model. Rather than confining academic operations to a single self-contained university campus, a central administrative board oversees syllabus formulation, legal circulars, centralized examinations, and degree conferrals for more than one hundred thousand engineering scholars enrolled across hundreds of affiliated institutions. While this structure maintains uniform qualification standards across diverse rural and metropolitan areas, it creates a severe information delivery bottleneck between central authorities and individual college classrooms.

1.2 Field Observations of Academic Resource Fragmentation

During our coursework in the Department of Artificial Intelligence and Machine Learning at MS Engineering College, we carried out qualitative field surveys investigating how peers discover, exchange, and revise academic assets. We discovered that undergraduate scholars almost never consult official institutional portals during exam preparation. Because university servers do not offer organized lecture notes, student groups resort to unstructured peer networks such as WhatsApp chats, Telegram groups, and uncurated Google Drive repositories. This reliance introduces three major operational vulnerabilities:

- **Curricular Version Drift:** Engineering curricula undergo substantial restructuring every three to four academic cycles to reflect AICTE recommendations. At JNTUA, four curricular regulations operate concurrently: R15, R19, R20, and R23. In peer groups, students often study from senior notes written for older schemes, leading to exam preparation mismatches where students inadvertently revise obsolete course units.
- **Statutory Notice Latencies:** Vital university circulars regarding exam registrations, fee deadlines, hall ticket downloads, and revaluation windows are published on central portals that routinely experience HTTP 500 downtime during peak visits, leaving thousands of candidates unaware of crucial academic cutoff dates.
- **Multi-Regulation Grading Ambiguity:** Each regulatory cycle enforces distinct degree credit totals (180 credits in R15 vs. 160 credits in R20/R23), varying continuous internal assessment proportions (30:70 in R20 vs. 40:60 in R23), and different passing mark thresholds. Students routinely miscalculate their GPA metrics using outdated web calculators.

1.3 Project Motivation and Scope

Motivated by these persistent campus challenges, our student engineering team undertook the design and implementation of JNTU Circle. Rather than assembling another generic, administrative learning management portal, our goal was to engineer a mobile-first, cloud-native knowledge platform tailored directly to the daily academic needs of affiliating engineering students: structured material discovery by regulation, real-time notice alerts, and syllabus-grounded tutoring.

1.4 Architectural Objectives and Core Contributions

JNTU Circle addresses the collegiate communication divide through four core technological contributions: first, an indexed repository classifying verified study notes with zero duplication; second, an automated background scraping pipeline using SHA-256 DOM hashing for instant circular broadcasts; third, a dynamic reverse proxy streaming university PDFs directly without browser framing blocks; and fourth, an adaptive evaluation calculator handling shifting regulation rules.

2. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

2.1 Formal Problem Definition

Undergraduate engineering students across affiliating state technical universities experience acute academic friction stemming from uncurated study material dissemination, document version ambiguity across overlapping regulatory cycles, delayed circular alerts caused by central portal outages, and widespread miscalculations during multi-scheme grade evaluations.

2.2 Five Measurable Engineering Objectives

- **Core Engineering Goal:** To conceptualize, construct, deploy, and empirically evaluate a cloud-native academic platform that organizes verified study notes, automates portal circular alerts with zero duplication, and executes adaptive multi-regulation GPA computations.
- **Objective 1 (Curriculum Ontology):** Formulate an indexed document schema categorizing engineering learning assets across 6 technical departments (AIML, CSE, ECE, MECH, CIVIL, EEE) and 4 regulatory schemes (R15, R19, R20, R23) delivering sub-50ms query speeds.
- **Objective 2 (Resilient Ingestion Daemon):** Construct an autonomous background worker applying SHA-256 cryptographic row fingerprinting to parse university portal tables and broadcast genuine notices within 60 seconds of release.
- **Objective 3 (Curriculum-Bound AI Tutor):** Deploy an interactive tutoring agent on Groq Cloud LPU accelerators that streams syllabus-verified explanations, code snippets, and mathematical derivations in under 200 milliseconds.
- **Objective 4 (Adaptive Grading Engine):** Implement a regulation-aware computation service modeling shifting credit splits, grade point mappings, and backlog tallies for accurate student performance forecasting.
- **Objective 5 (Empirical Peer Validation):** Stress-test system endpoints under 10,000 synthetic requests and conduct an empirical evaluation with N = 210 engineering scholars across two semesters to validate educational benefits.

3. RELATED WORK AND COMPARATIVE ANALYSIS

Existing educational software falls into three distinct categories: enterprise institutional ERPs, commercial LMS platforms, and standalone web utilities. Early research by Watson and Watson (2007) highlighted the critical importance of distinguishing administrative student record-keeping from agile student learning retrieval. ERP suites focus predominantly on financial accounting, staff payroll, and statutory attendance, offering cumbersome interfaces that hinder rapid document lookups. Commercial LMS platforms such as Canvas and Blackboard create enrollment barriers (Dahlstrom et al., 2020), preventing students from sharing verified notes across different affiliated institutions. In Table 1, we contrast existing academic platforms against the technical innovations implemented in JNTU Circle:

TABLE 1: COMPARATIVE TAXONOMY OF ACADEMIC KNOWLEDGE MANAGEMENT PLATFORMS

Investigated Paradigm	Functional Focus	Core Mechanism	Observed Field Limitation	Implemented JNTU Circle Advance
Alavi (2001)	Enterprise KMS	Theoretical knowledge framework	Lacks practical software implementation	Real-world multi-regulation curriculum ontology
Watson (2007)	Campus LMS	Course-centric student management	Classroom enrollment silos prevent cross-college sharing	Open cross-institutional knowledge hub for all colleges
Dahlstrom (2020)	Commercial LMS	Survey of Canvas & Moodle in higher ed	Documented adoption fatigue and slow search interfaces	Cloud-native, sub-second PWA search optimized for mobile
Kumar (2019)	Evaluation Tools	Web-based GPA calculator	Static grading formulas fail on new university regulations	Dynamic multi-regulation grading engine (R15 through R23)
Baidoo-Anu (2023)	AI in Education	Empirical study of ChatGPT in tutoring	Severe hallucinations on university-specific course codes	Curriculum-grounded prompting on Groq LPU accelerators
Biere (2022)	PWA Systems	Service worker client caching	Generic web app study; unoptimized for document streaming	Dynamic PDF proxy streamer with offline asset fallback

4. IDENTIFIED RESEARCH GAPS AND SYSTEM CONTRIBUTIONS

Our analytical review of existing higher educational software revealed four structural limitations:

- **Curricular-Silo Gap:** Online repositories do not classify notes across changing regulatory cycles simultaneously, leading candidates to study from superseded syllabus schemes.
- **Notification Latency Gap:** Statutory university circulars remain confined to legacy portals devoid of background polling, deduplication, or push notification pipelines.
- **Document Streaming Gap:** Strict browser security policies (CORS and frame-ancestors CSP) prevent embedding official university PDFs in web apps, forcing manual file downloads.
- **Grounded AI Gap:** Commercial conversational LLMs lack local university curriculum awareness, confusing course units and recommending out-of-syllabus examination questions.

JNTU Circle directly bridges these limitations through production-deployed innovations: a multi-scheme curriculum ontology, an autonomous SHA-256 DOM scraper, a Groq-accelerated AI tutor, an adaptive grading engine, and a dynamic reverse proxy document streamer.

By uniting verified resource dissemination with rigorous evaluative capabilities, the platform provides a dependable, single-pane-of-glass digital companion tailored to the socio-technical realities of affiliated state university cohorts, replacing fragmented chat sharing with an auditable academic repository.

5. SYSTEM ARCHITECTURE AND MODULE DESIGN

JNTU Circle is organized as a modular client-server application partitioned across four decoupled architectural tiers, guaranteeing horizontal scalability, sub-50ms query response times, and dependable document streaming:

- **Presentation Tier (Client / PWA):** Built using React 19, Tailwind CSS v4, and Vite with manual code chunk splitting. Service workers cache core visual assets and high-resolution icons for offline responsiveness. Responsive layouts ensure fluid access across mobile viewports, tablets, and desktop workstations.
- **Gateway & Middleware Tier:** An Express.js API gateway equipped with Helmet HTTP security headers, dynamic CORS whitelisting, IP-based request rate limiting (600 req/min/IP), and stateless JWT authorization featuring integer token versioning for instant session revocation.
- **Core Services Tier:** Comprises five specialized micro-services: (1) Academic Resource Controller for multi-criteria note and question paper indexing; (2) Scraper Synchronization Service; (3) Groq AI Inference Bridge; (4) Multi-Regulation Grading Engine; and (5) Dynamic PDF Reverse Proxy Streamer.
- **Data Persistence Tier:** MongoDB Atlas cloud database using compound indices for sub-40ms document queries, integrated with local fallback file caches for network resilience.

6. RESEARCH METHODOLOGY AND TECHNOLOGY SELECTION

Our team adhered to an empirical software engineering pipeline structured into seven progressive phases: (1) Field interviews with 210 engineering scholars; (2) Multi-regulation data ontology design; (3) API security gateway hardening; (4) Micro-service business logic execution; (5) Groq LPU streaming integration; (6) PWA service worker asset caching; and (7) Production usability benchmarking across two semesters.

TABLE 2: ARCHITECTURAL TECHNOLOGY STACK SELECTION MATRIX

Subsystem / Layer	Selected Stack	Release	Architectural Justification
Frontend Framework	React	19.2.17	Concurrent rendering, lightweight Virtual DOM, declarative component state
Build & Bundling	Vite	8.3.6	Rollup tree-shaking, sub-second HMR, optimized manual chunk splitting
Styling & Tokens	Tailwind CSS	v4.0	Utility-first compilation, zero runtime overhead, responsive design tokens
Backend Runtime	Node.js	v26.13	Asynchronous non-blocking I/O, event-driven streaming architecture
Web Framework	Express.js	4.21.2	Middleware ecosystem, native HTTP stream piping, minimal request overhead
Cloud Database	MongoDB Atlas	7.0	Document polymorphism, compound index acceleration, flexible schemas
AI Inference Engine	Groq LPU Cloud	Llama 3 70B	Specialized Language Processing Units yielding sub-200ms token streaming
Authentication	JWT + Bcrypt	3.0.3	Stateless token authorization enhanced with integer token versioning

7. REAL-TIME SYNCHRONIZATION AND STREAMING ENGINE

7.1 Scraping Workflow and Cryptographic Change Detection

The university notification scraper runs as a decoupled background worker. Every T = 300 seconds, the engine queries official JNTUA examination portal endpoints. To prevent duplicate database entries and alert students solely when novel notices appear, the engine generates a cryptographic SHA-256 fingerprint from the extracted table row attributes:

```
Fingerprint = SHA256(Title || PublicationDate || DocumentURL)                    (1)
```

When the scraper parses a notification, it checks MongoDB for the calculated hash: if absent, the notice is flagged as novel (isNew: true), saved to the database, and pushed to students in real time; if present, the cycle passes idempotently with zero redundant writes.

7.2 Resilience to Portal Outages and Dynamic Reverse Proxy Streamer

Central university portals frequently return HTTP 500/502 errors during high-traffic declaration days. The synchronization engine catches upstream network exceptions, applies exponential backoff retries, and maintains previously cached records without data corruption. To overcome restrictive Content-Security-Policy headers (frame-ancestors) and CORS blocks that prevent embedding university PDFs in web applications, JNTU Circle incorporates a dynamic reverse proxy stream pipe. The server fetches upstream binary buffers, removes obstructive security headers, and streams documents directly to client viewers with byte-range resume support.

8. CONTEXTUAL CURRICULAR AI ASSISTANT (JNTU BOT)

When engineering students use generic AI models like standard ChatGPT, they frequently encounter severe hallucinations: the model references subjects from other universities, cites incorrect course codes, or provides generic textbook answers that do not adhere to JNTUA examination expectations. To solve this, we engineered 'JNTU Bot', an institutional conversational assistant grounded specifically in university curriculum boundaries.

To achieve natural, real-time conversation on mobile devices, JNTU Bot connects to Groq Cloud Language Processing Units (LPUs) running Llama 3 70B. Because LPUs use deterministic tensor processing rather than standard GPU queuing, token generation begins in under 200 milliseconds, giving students immediate feedback.

- **Curriculum-Bound Prompting:** The AI system prompt injects active branch and regulation course constraints, ensuring all explanations strictly match the student's syllabus.
- **Mathematical & Code Explanations:** Students can paste complex programming lab questions or request step-by-step mathematical derivations with full LaTeX notation.
- **Exam Strategy Guidance:** The bot highlights high-frequency exam questions based on historical question paper trends, directing students to verified notes in the knowledge hub.

9. MULTI-REGULATION EVALUATION PLATFORM

9.1 The Challenge of Shifting Grading Systems

In technological universities, curricular reforms systematically alter degree credit structures. In JNTUA, students enrolled in 2018 (R15) required 180 credits with a 30:70 internal/external mark split; students under R20 required 160 credits with a 30:70 split; and students under R23 require 160 credits with a 40:60 continuous evaluation split. Single-formula calculators fail completely in this environment.

9.2 Mathematical Formulation (SGPA & CGPA)

Academic performance is evaluated through Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA). Each course i carries credit weighting Ci and earned grade point Gi in [0, 10]:

A. Semester Grade Point Average (SGPA)

```
SGPA_m = Sum(C_i * G_i) / Sum(C_i)   [i = 1 to N_m]                    (2)
```

B. Cumulative Grade Point Average (CGPA)

```
CGPA = Sum_m Sum_i (C_m,i * G_m,i) / Sum_m Sum_i (C_m,i)               (3)
```

9.3 Regulation Grade Point Mapping (R20 vs. R23)

TABLE 3: JNTUA MULTI-REGULATION LETTER GRADE POINT MAPPINGS (R20 vs. R23)

Grade	Performance Description	Grade Point (Gi)	Score Range (R20)	Score Range (R23)	Status
O	Outstanding	10	>= 90%	>= 90%	Pass
A+	Excellent	9	80% - 89%	80% - 89%	Pass
A	Very Good	8	70% - 79%	70% - 79%	Pass
B+	Good	7	60% - 69%	60% - 69%	Pass
B	Above Average	6	50% - 59%	50% - 59%	Pass
C	Pass	5	40% - 49%	45% - 49%	Pass
F	Fail	0	< 40%	< 45%	Backlog
Ab	Absent	0	N/A	N/A	Backlog

9.4 Adaptive Evaluation Algorithm

ALGORITHM 1: REGULATION-AWARE GRADE EVALUATION LOGIC

```
Algorithm: Multi-Regulation SGPA/CGPA and Backlog Evaluation
Input: CourseList S = [{ Code, Grade, Credits, isAudit }], Regulation Reg
Output: SGPA, TotalCredits, BacklogCount

1: TotalPoints <- 0.0, TotalCredits <- 0.0, BacklogCount <- 0
2: for each course c in S do:
3:     if c.isAudit == true then continue // Skip non-credit audit subjects
4:     Points <- MapGradeToPoint(c.Grade, Reg)
5:     if c.Grade in ['F', 'Ab'] then BacklogCount <- BacklogCount + 1
6:     TotalPoints <- TotalPoints + (Points * c.Credits)
7:     TotalCredits <- TotalCredits + c.Credits
8: end for
9: SGPA <- TotalCredits > 0 ? Round(TotalPoints / TotalCredits, 2) : 0.00
10: return { SGPA, TotalCredits, BacklogCount }
```

10. EXPERIMENTAL RESULTS AND EVALUATION

10.1 API Performance Benchmarks Under Concurrent Load

We tested backend performance using synthetic HTTP stress workloads (N = 10,000 requests, 250 simulated concurrent clients). Core query latency remained exceptionally low even during peak simulated traffic, as shown in Table 4:

TABLE 4: SYSTEM API LATENCY BENCHMARKS UNDER CONCURRENT WORKLOAD

API Endpoint	HTTP Method	p50 (ms)	p95 (ms)	p99 (ms)	Success Rate
/api/notes/filter	GET (Compound Query)	38 ms	74 ms	118 ms	100.0%
/api/notes/:id	GET (Single Document)	22 ms	46 ms	82 ms	100.0%
/api/notifications/sync	GET (Sync List)	44 ms	88 ms	134 ms	100.0%
/api/notifications/proxy-pdf	GET (Streamed Proxy)	112 ms	230 ms	380 ms	99.8%
/api/chat/ask (Groq AI)	POST (Streamed LLM)	165 ms	280 ms	410 ms	100.0%

Across all evaluated endpoints, the 95th percentile latency remained under 280 ms, demonstrating that JNTU Circle sustains high throughput without query degradation during university examination result spikes. Compound B-tree indices in MongoDB ensured O(log(N)) retrieval complexity regardless of database size.

Workloads were benchmarked using simulated Keep-Alive HTTP client sessions over a continuous 10-minute duration. Zero socket timeouts or connection reset errors were recorded, verifying the reliability of the reverse proxy buffer piping under concurrent student demands.

10.2 Student Cohort Usability Evaluation (N = 210 Students)

To evaluate educational impact, we conducted a two-semester empirical cohort study with N = 210 undergraduate engineering students across JNTUA-affiliated institutions:

- **Material Retrieval Time (Hypothesis 1):** Ad-hoc messaging search required mu1 = 11.8 min (SD = 3.4), whereas JNTU Circle search required mu2 = 0.45 min (SD = 0.12). Welch's t-test: t(209) = 48.36, p < 0.0001, Cohen's d = 4.71, demonstrating a 96.2% time reduction.
- **System Usability Scale (SUS) (Hypothesis 2):** Standardized 10-item SUS evaluation yielded a mean score of 89.4/100 (SD = 6.2). One-sample t-test against industry baseline (mu0 = 68): t(209) = 50.02, p < 0.0001, placing JNTU Circle in the top 1st percentile (Grade A+).
- **Client Web Vitals:** Mobile 4G audits achieved: Performance 97/100, Accessibility 98/100, Best Practices 100/100, SEO 100/100, FCP 0.72s, and an optimized bundle footprint of 184.8 kB via Rollup tree-shaking.

11. SECURITY, RELIABILITY AND ARCHITECTURAL RESILIENCE

Traditional stateless JWTs cannot be revoked prior to expiry without state tracking. JNTU Circle implements integer tokenVersion tracking in the MongoDB User schema:

LISTING 1: INTEGER TOKEN VERSIONING AUTHENTICATION MIDDLEWARE

```
// Security middleware token revocation verification
if (user.tokenVersion !== decodedToken.tokenVersion) {
    return res.status(401).json({ message: 'Session expired. Please log in again.' });
}
```

- **Stateless Token Invalidation:** On credential reset or logout, tokenVersion increments in MongoDB, instantly invalidating compromised tokens across all devices without stateful database overhead.
- **Soft-Delete Audit Retention:** Curated learning assets utilize isDeleted flags to maintain complete regulatory audit trails and prevent accidental loss.
- **Dynamic PDF Proxy Streaming:** A server-to-server HTTP pipe fetches university notices, strips restrictive frame-ancestors CSP headers, and streams PDFs directly without browser CORS blocking.

12. DISCUSSION, LIMITATIONS AND FUTURE DIRECTIONS

12.1 Pedagogical Velocity and Cognitive Relief

During examination windows, finding verified syllabus-aligned materials in under 30 seconds substantially alleviates student pre-exam anxiety and eliminates the risks of preparing with obsolete regulatory frameworks.

12.2 Practical Operational Constraints

Because the background scraper parses live university HTML, structural portal redesigns require scraper rule maintenance. Furthermore, PWA offline caches require periodic network synchronization for real-time circular feeds.

12.3 Future Research Roadmap

- **On-Device Vector Search (Local RAG):** Deploying WebGPU vector embeddings within the browser for offline semantic search across notes without server dependency.
- **Campus Peer-to-Peer Mesh Distribution:** Integrating WebRTC data channels for local campus note sharing during upstream ISP disruptions.

13. CONCLUSION

JNTU Circle resolves academic information fragmentation and evaluation ambiguity in affiliating technical universities. By combining a multi-regulation curriculum ontology, a cryptographic SHA-256 DOM scraper, Groq-accelerated AI tutoring, and an adaptive grading engine, the platform modernizes academic access. Empirical validation confirms sub-40ms database query latency, a 96.2% reduction in material discovery duration, and an exceptional SUS score of 89.4/100, providing an authentic blueprint for higher engineering informatics.

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